

AD ORAL QUESTIONS AND ANSWERS

1. Difference between distributed database and parallel database:

- Distributed Database: Data is spread across multiple nodes or locations, and each node can independently process queries and transactions. Coordination between nodes is crucial for consistency and reliability.
- Parallel Database: Processing tasks are divided among multiple processors or nodes, allowing for simultaneous execution of operations. This parallelism speeds up query processing and data manipulation tasks significantly.

2. Difficulties faced in transform operation:

Transform operations in ETL (Extract, Transform, Load) processes involve converting raw data into a usable format. Common difficulties include handling data inconsistencies, ensuring data quality, dealing with large volumes efficiently, and maintaining compatibility with target systems.

3. Issues faced in extraction phase:

Extraction involves retrieving data from various sources. Challenges often include dealing with heterogeneous data formats, managing data volume and velocity, ensuring data accuracy during transfer, and addressing potential source system downtime affecting extraction schedules.

4. Difference between object-oriented and object-relational type:

- Object-Oriented Database: Designed to store and manage objects as used in object-oriented programming. It supports concepts like inheritance, encapsulation, and polymorphism directly.
- Object-Relational Database: Combines relational database features with object-oriented principles. It extends the relational model to support complex data types, methods, and object-oriented concepts while maintaining compatibility with SQL.

5. NoSQL is what type of database:

NoSQL (Not Only SQL) encompasses various database technologies designed to handle large volumes of unstructured or semi-structured data. It includes document stores, key-value stores, column-family stores, and graph databases, among others.

6. Is a data warehouse needed for a data lake:

While a data lake can store vast amounts of raw, unstructured data, a data warehouse is typically needed for structured data, curated for specific analytical purposes. They serve different but complementary roles in an organization's data architecture, with the data lake acting as a source for raw data and the data warehouse providing organized, processed data for analytics.

7. HOLAP (Hybrid OLAP):

HOLAP combines features of both ROLAP (Relational OLAP) and MOLAP (Multidimensional OLAP) systems. It uses relational databases for storing and managing large datasets while employing multidimensional storage for analytical processing, striking a balance between flexibility and performance in OLAP operations.

8. Query for inheritance (SQL):

In SQL, object-oriented concepts like inheritance are not directly supported. Instead, you can model inheritance relationships using tables and relationships. For example:

```
```sql
```

```
CREATE TABLE Person (
 id INT PRIMARY KEY,
 name VARCHAR(50)
);
```

```
CREATE TABLE Employee (
 emp_id INT PRIMARY KEY,
 salary DECIMAL(10, 2),
 dept_id INT,
 FOREIGN KEY (emp_id) REFERENCES Person(id)
);
```
```

9. Example of in-memory:

An example of in-memory computing is using RAM (Random Access Memory) to store and process data directly in memory, bypassing traditional disk-based storage. In-memory databases like Redis or Memcached excel at rapid data access and processing, making them ideal for caching, real-time analytics, and high-performance applications.

10. Data mart definition:

A data mart is a subset of a data warehouse focused on a specific business area or department. It contains summarized, relevant data tailored to the needs of a particular user group, enabling efficient and targeted analytics and reporting.

11. Why use a relational database:

Relational databases offer structured data storage with well-defined schemas, ACID (Atomicity, Consistency, Isolation, Durability) transactions ensuring data integrity, SQL support for complex queries, and efficient indexing for fast data retrieval, making them suitable for various transactional and analytical tasks.

12. Data cleaning definition:

Data cleaning involves identifying and correcting errors, inconsistencies, and missing values in datasets to improve data quality and accuracy. It includes tasks like deduplication, outlier detection, standardization, and validation.

13. Data types in spatial databases:

Spatial databases support specialized data types for storing spatial/geographical information, such as points, lines, polygons, and geometries. These data types enable spatial indexing and spatial operations like distance calculations, containment checks, and spatial joins.

14. Uses of spatial data:

Spatial data is used in geographic information systems (GIS), mapping applications, urban planning, environmental monitoring, logistics, location-based services, and more. It helps analyze relationships between geographical entities, visualize data spatially, and derive insights from geographic patterns.

15. Intra-query parallelism:

Intra-query parallelism involves parallel execution of tasks within a single query or operation. Parallel databases use techniques like parallel query processing, data partitioning, and parallel join algorithms to distribute workload among multiple processors or nodes, speeding up query execution.

16. Type inheritance in SQL and syntax:

SQL doesn't directly support type inheritance like object-oriented languages. In SQL databases, you typically model inheritance using relationships between tables. For example:

```
```sql
```

```
CREATE TABLE Vehicle (
 id INT PRIMARY KEY,
 type VARCHAR(50)
);
```

```
CREATE TABLE Car (
 car_id INT PRIMARY KEY,
 color VARCHAR(20),
 FOREIGN KEY (car_id) REFERENCES Vehicle(id)
);
...
```

**17. Concurrency control and recovery in parallel databases:**

Parallel databases employ techniques like locking, timestamping, and multiversion concurrency control (MVCC) to manage concurrent access and ensure data consistency. Recovery mechanisms include logging changes, checkpoints, and transaction rollback/commit protocols to maintain database integrity during failures.

**18. Relation between NoSQL and spatial data:**

NoSQL databases can store and handle spatial data using specialized data types or extensions. For example, document-oriented databases like MongoDB can store GeoJSON objects, graph databases can represent spatial relationships efficiently, and column-family stores can manage spatial data attributes alongside other data.

**19. Data mart definition (repeated):**

A data mart is a subset of a data warehouse focused on a specific business area or department. It contains summarized, relevant data tailored to the needs of a particular user group, enabling efficient and targeted analytics and reporting.

**20. Example of ORDBMS in spatial databases:**

PostgreSQL with its PostGIS extension is a prime example of an Object-Relational Database Management System (ORDBMS) used in spatial databases. PostGIS adds spatial data types, functions, and indexing capabilities to PostgreSQL, making it suitable for spatial data storage and analysis.

**21. Difference between data lakes and data warehouses:**

Data lakes store raw, unstructured, and semi-structured data at scale, providing a centralized repository for diverse data sources. Data warehouses, on the other hand, store structured and processed data optimized for analytics and reporting, supporting OLAP queries and business intelligence tasks.

**22. Efficiency of parallel databases in query running:**

Parallel databases achieve query efficiency through parallel execution of tasks across multiple processors or nodes. They leverage parallel query processing, data partitioning, indexing strategies, and optimized join algorithms to reduce response times for complex queries and large datasets.

**23. Snowflake and Star Schema Data warehouse, Factless table meaning:**

Snowflake and Star Schema are data modeling techniques used in data warehousing.

-

Star Schema: It features a central fact table surrounded by dimension tables, forming a star-like structure. Fact tables contain numeric measures, and dimension tables provide context with descriptive attributes.

- Snowflake Schema: Similar to Star Schema but with normalized dimension tables, reducing data redundancy. It leads to more normalized data storage but can increase query complexity.
- Factless Table: A fact table without any measures, often used to represent relationships between dimensions or events where measures are not applicable.

#### **24. Father of Data warehouse definition:**

Bill Inmon is often considered the "father of data warehousing" for his pioneering work in developing concepts and methodologies for building data warehouses in the 1980s.

#### **25. Can relational databases be replaced by object databases? Why or why not:**

Relational databases and object databases serve different purposes and have distinct strengths. Relational databases excel at structured data management, SQL queries, and ACID transactions, making them suitable for transactional systems. Object databases, on the other hand, are designed for handling complex, interconnected data structures used in object-oriented programming. They can coexist or integrate in systems based on specific needs, but outright replacement depends on the application's requirements.

#### **26. Data warehouse exact definition:**

A data warehouse is a centralized repository that integrates data from multiple sources, transformed into a consistent format suitable for analysis and reporting. It supports OLAP operations, historical data storage, and decision-making processes in organizations.

#### **27. Object database working:**

Object databases store and manage complex data structures directly, without the need for mapping to relational schemas. They support object-oriented concepts like inheritance, encapsulation, and polymorphism natively, making them suitable for object-oriented programming paradigms and applications.

#### **28. Time variant, subject-oriented property of data warehouse:**

- Time Variant: Data warehouses store historical data over time, allowing analysis of trends, patterns, and changes over specific periods.
- Subject-Oriented: Data in a warehouse is organized around specific business subjects or areas, facilitating focused analysis and reporting tailored to user needs.

#### **29. Explain OLAP types (ROLAP, MOLAP, HOLAP):**

- ROLAP (Relational OLAP): Uses relational databases for OLAP operations, leveraging SQL queries and relational model features.
- MOLAP (Multidimensional OLAP): Stores data in multidimensional arrays (cubes), optimized for fast analytical queries and complex calculations.
- HOLAP (Hybrid OLAP): Combines ROLAP and MOLAP features, using relational databases for storage and multidimensional structures for analysis, balancing flexibility and performance.

### **30. What is Metadata?**

Metadata refers to data that provides information about other data. It describes the structure, format, context, and relationships of data elements within a dataset or database. Metadata is crucial for data management, governance, and understanding data lineage and semantics.

### **31. OLAP vs OLTP (difference):**

- OLAP (Online Analytical Processing): Optimized for complex analytical queries, data aggregation, and reporting in decision support systems. Focuses on historical data analysis.

- OLTP (Online Transaction Processing): Optimized for transactional operations involving data insertion, updating, and retrieval in real-time. Focuses on current data for operational tasks.

### **32. P in CAP theorem, junk dimensions, aggregate dimensions, ETL process, queries for distributed databases:**

- P in CAP Theorem: Stands for "Partition tolerance," one of the three properties (along with Consistency and Availability) that distributed systems must balance according to the CAP theorem.

- Junk Dimensions: Combine several low-cardinality flags or indicators into a single dimension table to reduce dimensionality in data warehouses.

- Aggregate Dimensions: Dimensions derived from aggregating or summarizing data from other dimensions or fact tables, useful for pre-aggregated queries in OLAP systems.

- ETL Process: Stands for Extract, Transform, Load, the process of extracting data from source systems, transforming it into a suitable format, and loading it into a target system like a data warehouse.

- Queries for Distributed Databases: Distributed databases support queries involving distributed data across multiple nodes or locations, often requiring distributed query processing techniques and optimization strategies.